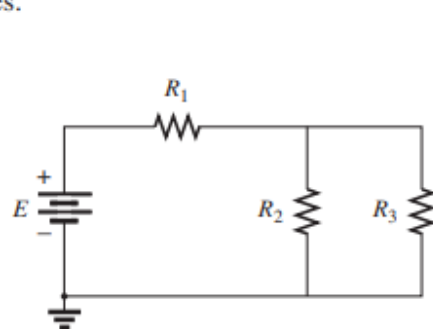
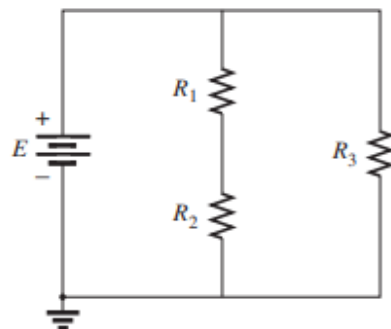


HW4

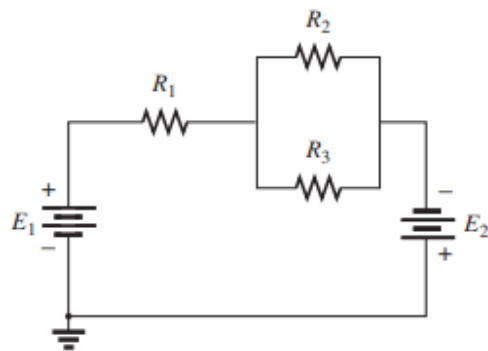
1. For each configuration in Fig. 5.87, find the individual (not combinations of) elements (voltage sources and/or resistors) that are in series.



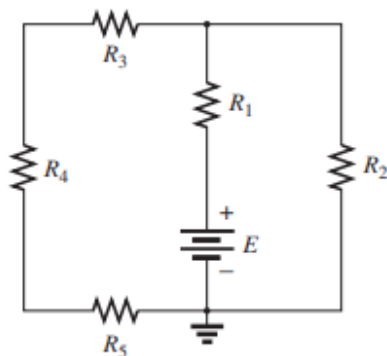
(a)



(b)



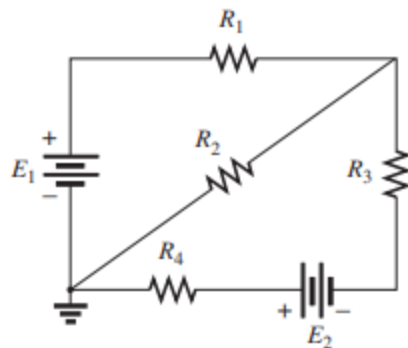
(c)



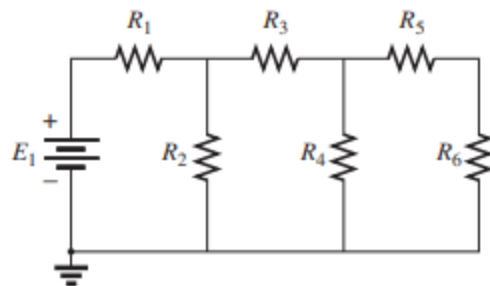
(d)

FIG. 5.87
Problem 1.

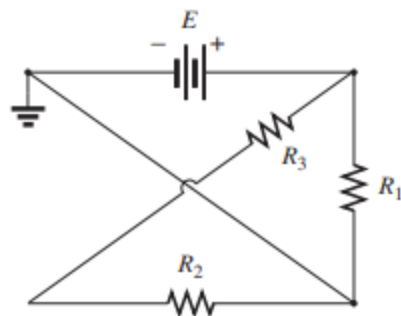
2. For each configuration in Fig. 5.88, find the individual (not combinations of) elements (voltage sources and/or resistors) that are in series.



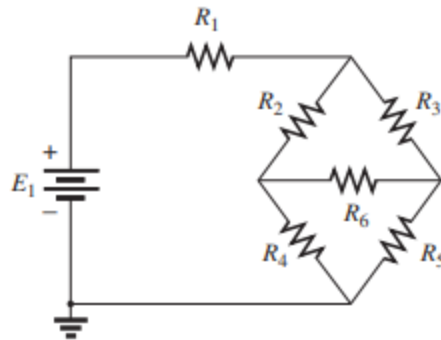
(a)



(b)



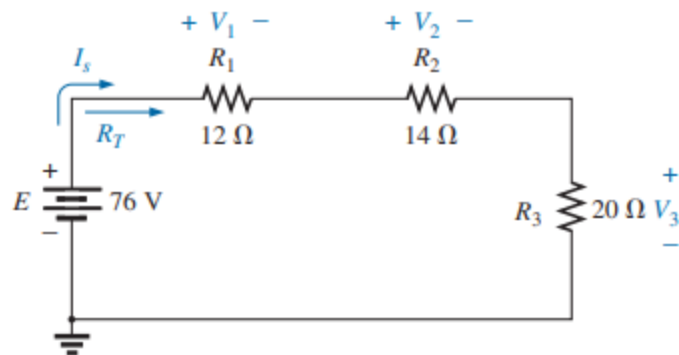
(c)



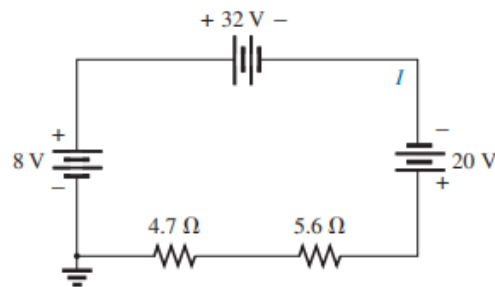
(d)

10. For the series configuration in Fig. 5.96, constructed of standard values:

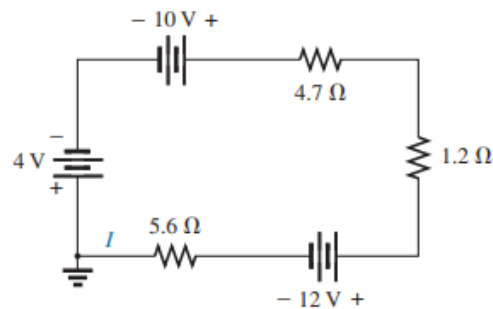
- Find the total resistance.
- Calculate the current.
- Find the voltage across each resistive element.
- Calculate the power delivered by the source.
- Find the power delivered to the $20\ \Omega$ resistor.



22. Determine the current I and its direction for each network in Fig. 5.108. Before solving for I , redraw each network with a single voltage source.

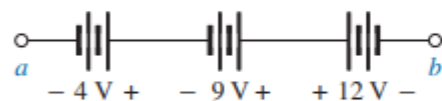


(a)

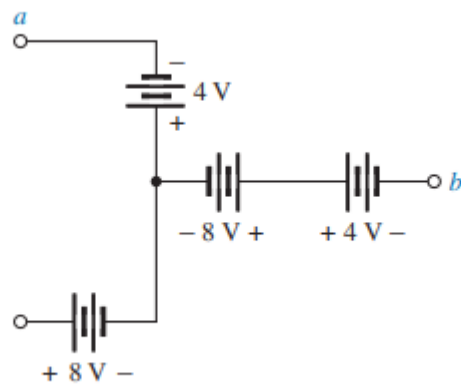


(b)

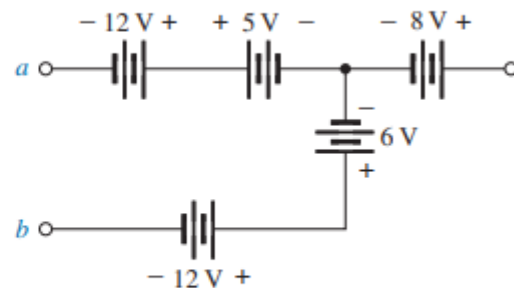
21. Combine the series voltage sources in Fig. 5.107 into a single voltage source between points a and b .



(a)

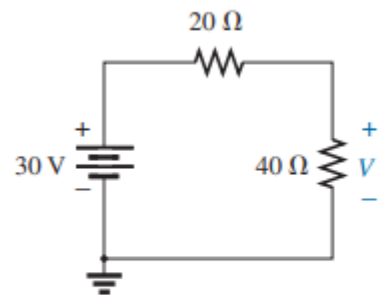


(b)

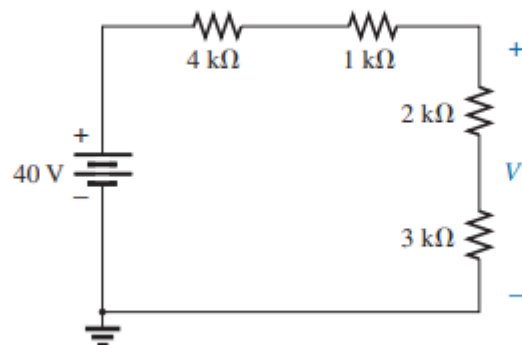


(c)

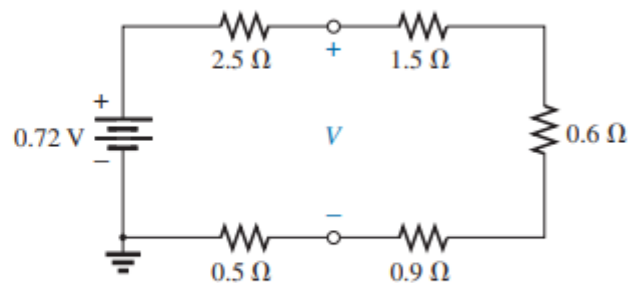
30. Using the voltage divider rule, find the indicated voltages in Fig. 5.116.



(a)

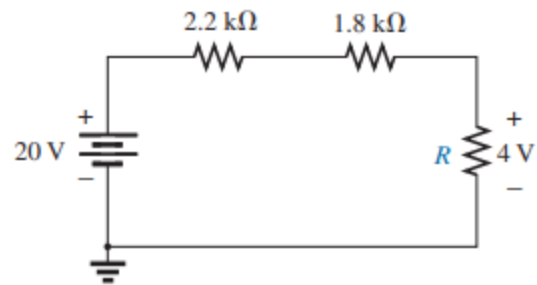


(b)

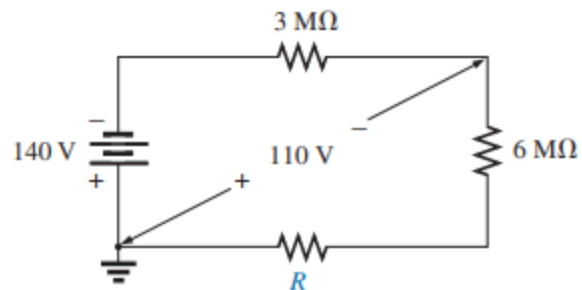


(c)

- *34. Using the voltage divider rule, find the unknown resistance for the configurations in Fig. 5.120.

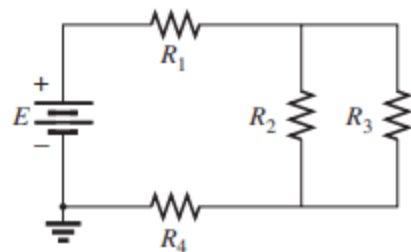


(a)

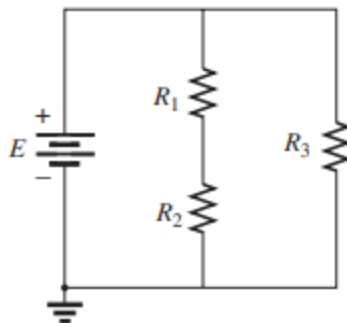


(b)

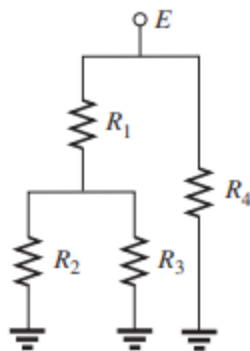
1. For each configuration in Fig. 6.61, find the voltage sources and/or resistors elements (individual elements, not combinations of elements) that are in parallel.



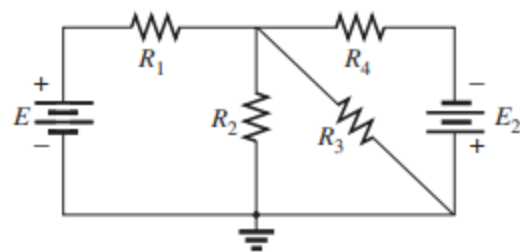
(a)



(b)

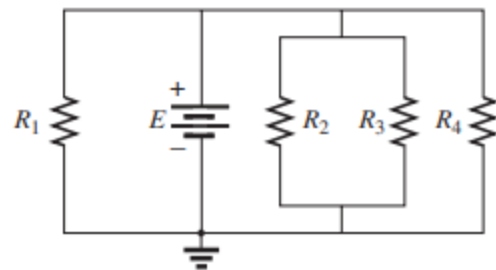


(c)

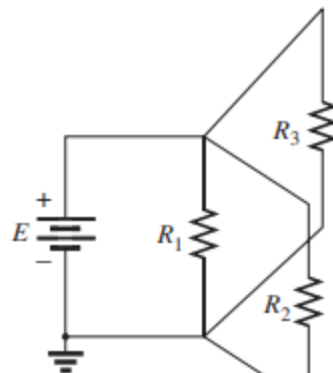


(d)

2. For each configuration of Fig. 6.62, find the voltage sources and/or resistive elements (individual elements, not combinations of elements) that are in parallel.



(a)



(b)

